

UNIT – I

Introduction to Advance Database Systems: Overview of advance database systems, their importance and Applications; **EER Model** -The ER model revisited, EER model: Super classes, Subclasses, Inheritance, Specialization and Generalization, Constraints and characteristics of specialization and Generalization, Category.

Object Model: Overview of Object-Oriented concepts, Object identity, Object structure, Type constructors, Encapsulation of operations, Methods, and Persistence, Type hierarchies and Inheritance, Complex objects, Schema design for OODBMS, OQL, Persistent Programming language, OODBMS architectures and storage issues, Transaction and concurrency control.

Object Relational Database and Information Retrieval: Database design for an ORDBMS – Nested relations and collections; Storage and access methods, Query processing and Optimization, Advance Querying: User define data types, manipulating objects table, object views; Information Retrieval & ways to retrieve information.

UNIT - II

Parallel Database: Architectures for parallel databases, Inter and Intra Query parallelism, Inter and Intra Query operations, Parallelizing individual operations, Sorting, Joins, Pipelining.

Distributed Database: Architectures for Distributed Database, Data Fragmentation, Replication, and Allocation Techniques for Distributed Database Design, Query processing in Distributed Databases; Concurrency Control and Recovery in Distributed Databases.

Overview of Client Server Architectures: Centralized and Client-Server architectures, Server architectures.

UNIT-III

Enhanced Data Models for Advanced Applications: Active database- syntax and semantics (DB2, Oracle), applications, design principles for active rules, Temporal database

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concepts, Spatial databases, Deductive databases.

Emerging Database Technologies: Mobile databases, Multimedia Databases, Geographic Information systems (GIS); XML and Internet Databases: Structured, Semi-structured and Unstructured Data, Introduction to web databases and XML, Structure of XML data.

UNIT – IV

Data Warehouse and OLAP Technology: Need for data warehouse, Definition, Goals of data Warehouse, Challenges faced during Warehouse Construction, Advantages, Types of Warehouse: Data Mart, Virtual Warehouse and Enterprise Warehouse; Components of Warehouse: Fact data, Dimension data, Fact table and Dimension table, Designing fact tables; Pre-requisite Phases: Extract, Transform and load process; Warehouse Schema: star, snowflake and galaxy schemas; OLTP vs OLAP, Strengths of OLAP, Applications of OLAP.

Multidimensional data models: Data Cubes & Data Cuboids, Lattice; OLAP operations: Advantages, Types: Roll up, Drill down, Pivot, Slice & Dice operations, Applications; OLAP Server: Need, Types: ROLAP, MOLAP and HOLAP, Features; Data Warehouse Implementation, Introduction to Efficient computation of data cubes.

MCA 2nd Semester 2 Year Course
(w.e.f. 2020-21) Examination, May-2023
**ADVANCE DATABASE SYSTEMS & DATA
WAREHOUSE**

Paper-20MCA22C1

Time allowed : 3 hours]

[Maximum marks : 80

Note : Attempt five questions in all. Question No.1 is compulsory. In addition to compulsory question, attempt four more questions selecting one question from each unit.

1. Compulsory Question

- (a) How category is different from shared subclass?
- (b) What is linear recursion in ORDBMS?
- (c) Name the partitioning techniques used in I/O parallelism.
- (d) Name the factors that do not appear in centralized systems that affect concurrency control and recovery in distributed systems.
- (e) How the delete and update commands would be implemented on valid time and bitemporal relations?
- (f) What is DTD in XML?
- (g) Differentiate between virtual and enterprise warehouse.

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- (h) Why data cubes are used in multidimensional models?

Unit-I

2. (a) Discuss various constraints applied on specialization and generalization. Also discuss how multiple inheritance can be achieved in case of specialization.
- (b) Differentiate between structured and unstructured complex objects. Also discuss reference semantics exist between a complex object and its components at each level.
3. (a) How query is processed and optimized in ORDBMS? Give an example also.
- (b) How effectiveness of information retrieval is measured? Discuss the metrics for measuring retrieval effectiveness.

Unit-II

4. (a) What is intraoperation parallelism? How parallelism can be increased with intraoperation parallelism?
- (b) What are various design issues related to DDBMS? Discuss their usage also.
5. (a) How server provides transaction services to client? Illustrate with diagrammatical notation.

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- (b) How concurrency control is achieved in distributed database?

Unit-III

6. (a) How time is incorporated using tuple versioning and attributes versioning in temporal database?
- (b) Discuss the architecture for mobile computing with diagram.
7. (a) How rules are interpreted in deductive database? Also discuss the datalog program and their safety.
- (b) What are different components and constraints associated with GIS?

Unit-IV

8. (a) What is data warehouse? Discuss different schema of data warehouse. From these schemas, which one is better and how?
- (b) How efficient computation of data cubes is specified in multidimensional model?
9. (a) Explain the ETL process with diagrammatic notation.
- (b) What is OLAP? Discuss various OLAP operation with example.

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9. (a) What is star and snowflake schema ? Explain with the help of example.
- (b) What is OLAP ? Discuss the different types of OLAP operations.

- (d) What is transparency feature of DDBMS ?
- (e) What is structured and semi structured data ?
- (f) What is temporal database ?
- (g) What is data cube ?
- (h) What is OLTP ?

UNIT – I

2. (a) What are the enhancements that distinguish the EER model from the ER model ?
- (b) What is the difference between specialization and generalization ? Why do we not display this difference in schema diagram ?
3. (a) Differentiate the following with respect to the object-oriented data model :
- (i) Objects and attributes
- (ii) Classes, subclasses and superclasses
- (b) What are the different ways of transaction management and concurrency control in OODBMS ?

UNIT – II

4. What is parallel database ? Explain various architectures of parallel databases along with their advantages and disadvantages.

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MCA 2nd Semester 2yr. Course w. e. f. 2020-21

Examination – July, 2022

**ADVANCE DATABASE SYSTEMS & DATA
WAREHOUSE**

Paper : 20MCA22C1

Time : Three hours]

[Maximum Marks : 80

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt five questions in all. Question Number 1 is compulsory. In addition to compulsory question, student has to attempt four more questions selecting one question from each Unit. All questions carry equal marks.

1. (a) What is object identity ?
- (b) What is type inheritance ?
- (c) What is difference Inter and Intra query parallelism ?

5. What is distributed database ? Explain following in the context of distributed databases :

- (i) Data Fragmentation
- (ii) Data Replication
- (iii) Data allocation techniques
- (iv) Concurrency control

UNIT – III

6. Explain in detail the following with their advantages and applications :

- (a) Mobile databases
- (b) Multimedia databases

7. (a) What is XML ? What are its advantages ? Explain the three main types of XML documents.

- (b) What is deductive database ? How rules are interpreted in deductive database ?

UNIT – IV

8. What is data warehouse ? Explain and compare following types of data warehouse :

- (i) Data Mart
- (ii) Virtual Warehouse
- (iii) Enterprise Warehouse

